

Performance Testing

Date	30 June 2026
Team ID	
Project Name	Rising Waters
Maximum Marks	5 Marks

Performance testing evaluates how the Vanguard HDI system behaves under expected and peak load conditions. The sections below document the testing approach, tools used, and results obtained.

Step 1: Testing Overview

Testing Tool Used	Python time profiling, unittest
Type of Testing	Load & Inference Speed Benchmarking
Target Module / API	Flask API Route /api/predict
Test Environment	Local development server
Test Date	30 June 2026

Step 2: Test Scenarios

#	Test Scenario	Virtual Users	Duration (sec)	Expected Outcome
1	Single simulation run	1	10	Calculates HDI score in < 30ms
2	Concurrent simulations	10	30	Processes all in < 50ms average
3	Rapid slider updates	50	60	Throughput > 100 req/sec, 0% errors

Step 3: Performance Test Results

#	Metric	Target	Actual	Status	Remarks
1	Response Time (Avg)	< 2 seconds	24ms	Pass	Calculates HDI instantly
2	Response Time (Max)	< 5 seconds	42ms	Pass	Under peak simulated concurrency
3	Throughput (Req/sec)	-	req/s	Pass	Zero-throttling on server
4	Error Rate	< 1%	0%	Pass	No failed calculations
5	CPU Utilization	< 80%	15%	Pass	Low CPU usage on development PC
6	Memory Utilization	< 80%	45MB	Pass	Light memory usage in Flask

Step 4: Observations & Analysis

Key Findings	Inference latency remains well under the 2-second target even under simulated concurrent load, confirming the Linear Regression model and Flask routing are lightweight enough for real-time use.
Bottlenecks Identified	None observed at the tested concurrency levels; SQLite write serialization is a potential future bottleneck under heavier multi-analyst load.
Optimization Steps Taken	Model kept lightweight (Linear Regression) and loaded once in memory at server start to avoid repeated disk reads on each request.